MLT Week 9

Perceptron Learning Algorithm

Discriminative Model for Classification

How to model $P(y = 1/x)$?

Simplest assumption: $P(y = 1/x) = \begin{cases} 1 & \text{if } w^T x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

$\therefore$ Data is Linearly Separable.

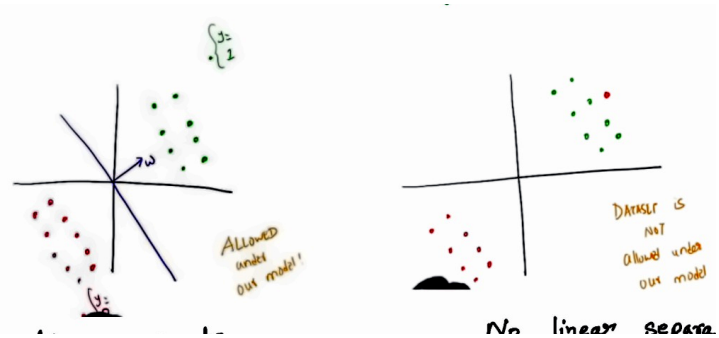


Figure 1. Two scatter plots — one with a clear linear separator (left, "Linear separate exists"), the other with overlapping clusters (right, "No linear separator").

Can we develop algorithms with the assumption that the data is linearly separable?

Goal: $\min_{h \in H_{\text{linear}}} \sum_{i=1}^{n} I(h(x_i) \neq y_i)$

$$H_{\text{linear}} = \{h_w : h_w(w) = \text{sign}(w^T x)\}$$

assuming "Linear Separability"

Perceptron Algorithm

Input: $\{(x_1, y_1), \ldots, (x_n, y_n)\}, \quad x_i \in \mathbb{R}^d, \quad y_i \in \{+1, -1\}$

$i = 1, \ldots, D, (x_1, y_1), \ldots, (x_n, y_n), w_i \in \mathbb{R}$

Iteration: $w^0 = 0 \in \mathbb{R}^d$

$$Y_i \in \{+1, -1\}, \qquad [0\ 0\ 0\ \ldots\ 0]$$

until convergence, - Pick (x_i, y_i) pair from the dataset - If sign$(w^T x_i) = $ sign(y_i), do nothing else, $w^{t+1} = w^t + x_i y_i$ ← update rule

Understanding the Perceptron Update Rule

If mistake, then $w^{t+1} = w^t + x_i y_i$

Mistake 1

Predicted: +1, Actual: -1, $y_i = -1$

$$w^{t+1} = w^t + x_i y_i$$

$$(w^{t+1})^T x_i = (w^t + x_i y_i)^T x_i = w^T x_i + y_i \|x_i\|^2 \geq 0 \ (> 0) \rightarrow \text{Negative}$$

Mistake 2

Predicted: -1, Actual: +1, $y_i = +1$

$$(w^{t+1})^T x_i = w^T x_i + y_i \|x_i\|^2 < 0 \ (> 0)$$

The update rule pushes w in the right direction for x_i.

Example 1: $w^{new} = w^{old} + x_i \frac{y_i}{y}$

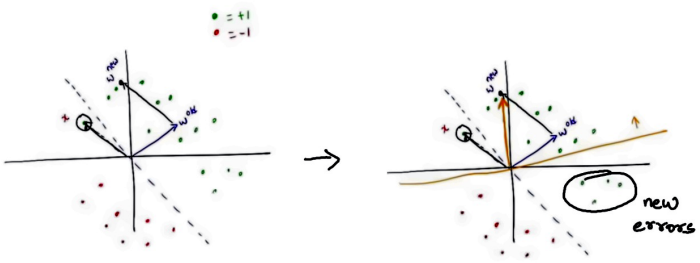


Figure 2. Two scatter plots of data points with a dashed decision boundary line, showing an initial misclassified point near the origin being corrected by the update rule, resulting in a new, more accurate boundary.

- Fixing w for one x, might affect decision for other data points.

Proof of Convergence of Perceptron Algorithm

We need more careful argument for convergence.

Example 2

Is this a linearly separable dataset?

$$\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, +1 \ ; \ \begin{bmatrix} 0 \\ -1 \end{bmatrix}, +1 \ ; \ \begin{bmatrix} -1 \\ 1/2 \end{bmatrix}, -1 \right\}$$

Is there a $w \in \mathbb{R}^2$ st $w^T x_i > 0 \Rightarrow y_i = +1, w^T x_i < 0 \Rightarrow y_i = -1$?

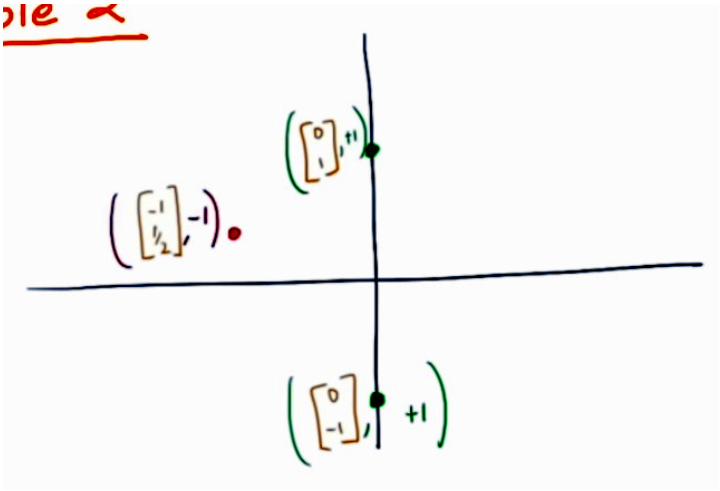


Figure 3. A line through the origin with a point $(\begin{bmatrix}1\\1/2\end{bmatrix}, -1)$ on one side labeled $+1$, and $(\begin{bmatrix}-1\\0\end{bmatrix}, +1)$, $(\begin{bmatrix}0\\-1\end{bmatrix}, +1)$ on the other side.

There is

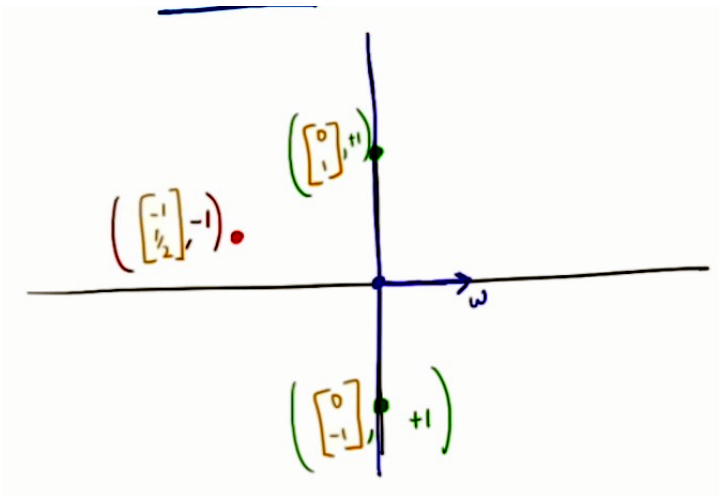


Figure 4. Same line through the origin, but now with a point $(\begin{bmatrix}1\\1/2\end{bmatrix}, -1)$ on the "wrong" side (below the line) labeled $+1$, and $(\begin{bmatrix}-1\\0\end{bmatrix}, +1)$, $(\begin{bmatrix}0\\-1\end{bmatrix}, +1)$ still on the "correct" side above the line.

Perceptron

$$w^0 = \begin{bmatrix}0\\0\end{bmatrix}, \quad w^{0T}x_1 = 0, \quad w^{0T}x_2 = 0, \quad w^{0T}x_3 = 0, \quad \hat{y}_1 = +1, \quad \hat{y}_2 = +1, \quad \hat{y}_3 = +1$$

$$w^1 = w^0 + x_3 y_3 = \begin{bmatrix}0\\0\end{bmatrix} + \begin{bmatrix}-1\\1/2\end{bmatrix} \times -1$$

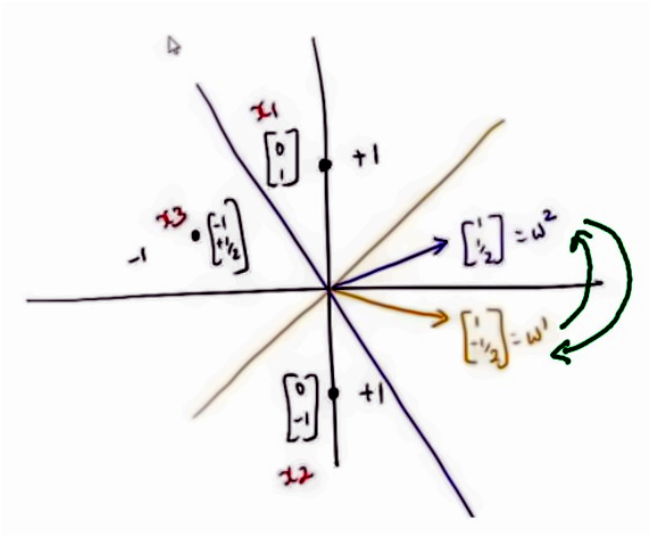


Figure 5. A scatter plot of points $\begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}-1\\1/2\end{bmatrix}, \begin{bmatrix}-1\\-1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix}$, all labeled $+1$. A separating line through the origin is shown, with an arrow from $\begin{bmatrix}-1\\1/2\end{bmatrix}$ to $\begin{bmatrix}1\\1/2\end{bmatrix} = w^*$, and another arrow from $\begin{bmatrix}-1\\-1\end{bmatrix}$ to $\begin{bmatrix}1\\-1\end{bmatrix} = w^*$.

$$w^1 = w^0 + x_3 y_3 = \begin{bmatrix}0\\0\end{bmatrix} + \begin{bmatrix}-1\\1/2\end{bmatrix} \times 1 = \begin{bmatrix}-1\\1/2\end{bmatrix}$$

$$w^2 = w^1 + x_1 y_1 = \begin{bmatrix}-1/2\\1/2\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} \cdot 1 = \begin{bmatrix}-1/2\\1/2\end{bmatrix}$$

$$w^3 = w^2 + x_2 y_2 = \begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix} \cdot 1 = \begin{bmatrix}1\\1\end{bmatrix}$$

$$w^3 = \begin{bmatrix}1\\-1/2\end{bmatrix} = w^1$$

We see that we are right back where we started. The algorithm is unable to provide a w which can correctly classify all the data points correctly. It just won't converge. Let's modify our assumption.

Assumption

Linear Separability with δ-Margin

A dataset $\{(x_1, y_1), \ldots (x_n, y_n)\}$ is linearly separable with γ-margin if $\exists w \in \mathbb{R}^d$ st $(w^T x_i) y_i \geq \gamma \ \forall i$, some $\gamma > 0$.

Proof of Convergence of Perceptron Algorithm

Radius Assumption

$\forall i \in$ Dataset, $\|x\|_2 < R$, for some $R > 0$

Without loss of generality, assume $\|w^*\| = 1$

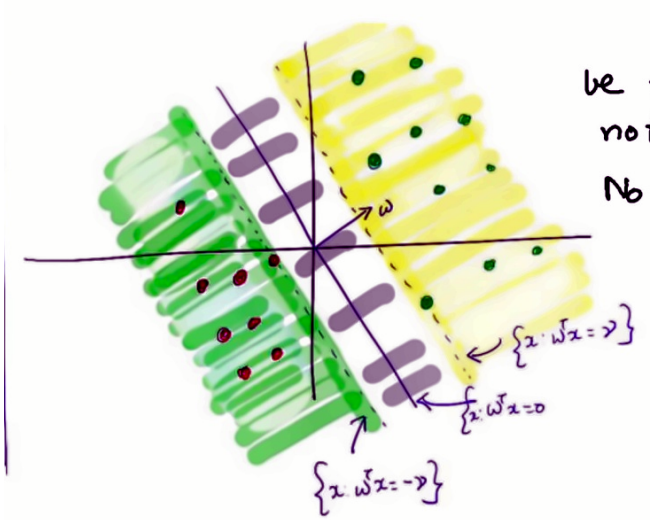


Figure 6. Two overlapping green and yellow regions separated by a margin band, labeled $\{x : w^T x \geq \gamma\}$ and $\{x : w^T x \leq -\gamma\}$, with a separating line w through the origin, and a shaded gap between them.

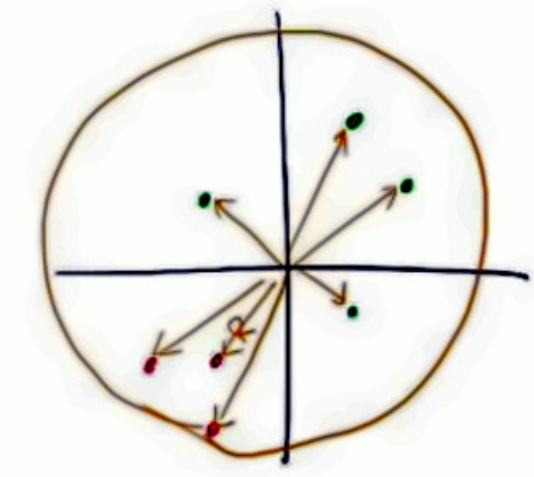


Figure 7. A circle of radius R centered at the origin with several data points (green crosses and red dots) inside it, each labeled with an arrow pointing from the origin.

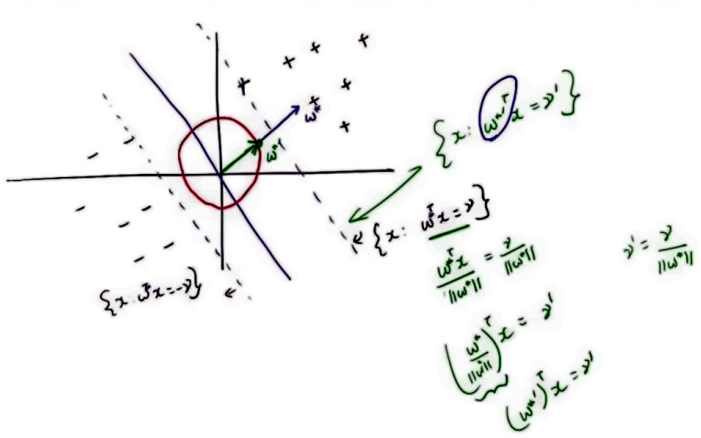


Figure 8. A dot product decision boundary around the origin — a circle of radius $y'/\|w^*\|$ — with a green vector w^* inside it, and two sets of data points ($+$'s and $-$'s) on either side of the boundary.

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\{x: w^*^Tx = y'\} \qquad \{x: w^Tx = y\}
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\frac{w^*^Tx}{\|w^*\|} = \frac{y}{\|w^*\|}, \qquad y' = \frac{y}{\|w^*\|}
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Warning: equation could not be typeset; shown as written.

$$\boxed{(w^*)^T x = y'} \qquad \boxed{(w^*)^T x = y}$$

Analysis of "Mistakes" of Perceptron

- Observe that an update happens only when a mistake occurs.
- Let w^l be the current w & a mistake occurs in wrt (x, y)

$$\therefore w^{l+1} = w^l + x \cdot y$$

$\|w^{l+1}\|^2 = \|w^l + xy\|^2 = (w^l + xy)^T(w^l + xy) = \|w^l\|^2 + 2(w^T x)y + \|x\|^2 y^2 \leq 0$ because mistake

$$\begin{aligned}
\|w^{l+1}\|^2 &\leq \|w^l\|^2 + R^2 \\
\|w^{l+1}\|^2 &\leq (\|w^{l-1}\|^2 + R^2) + R^2 \\
&\vdots \\
\|w^{l+1}\|^2 &\leq \|w^0\|^2 + lR^2
\end{aligned}$$

$$\boxed{\|w^{l+1}\|^2 \leq lR^2} \quad \text{(no of mistakes)} \qquad (\square)$$

$$(w^{l+1})^T w^* = (w^l + xy)^T w^* = w^T w^* + (w^{*T} x)y \geq \gamma$$

$$(w^{l+1})^T w^* \geq w^T w^* + \gamma$$
$$\geq (w^{l-1} w^* + \gamma) + \gamma$$

$$(w^{l+1})^T w^* \geq (w^*)^T w^* + \ell\delta$$

$$\boxed{(w^{l+1})^T w^* \geq \ell\delta}$$

For any (x, y),

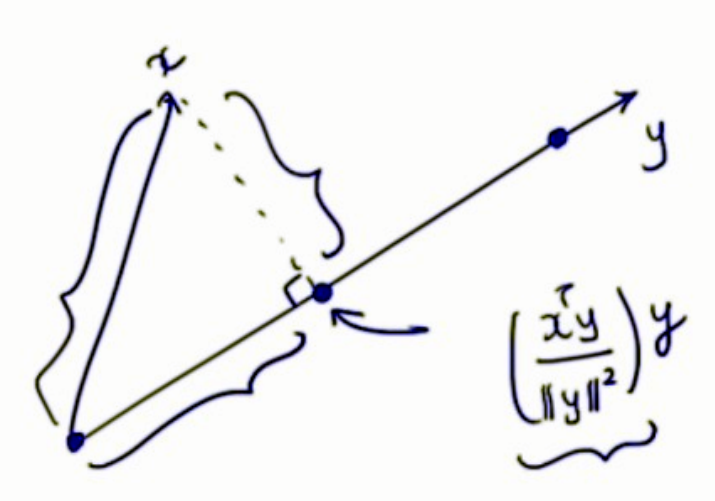


Figure 9. A point y on the axis with a dashed line to an off-axis point labeled x, and a dot at the origin.

$$\left\| \frac{x^T y}{\|y\|^2} y \right\|^2 \leq \|x\|^2 \quad \text{[Pythagoras]}$$

$$(x^T y)^2 \frac{\|y\|^2}{\|y\|^4} \leq \|x\|^2$$

Cauchy-Schwarz Inequality

$$(x^T y)^2 \leq \|x\|^2 \|y\|^2$$

From □:

$$\ell^2 \leq (w^{l+1})^T w^*$$
$$\Rightarrow \ell^2 \leq ((w^{l+1})^T w^*)^2 \leq \|w^{l+1}\|^2 \|w^*\|^2 \quad \text{[from C-S]}$$

$$\Rightarrow \|w^{l+1}\|^2 \geq \ell^2$$

$$\|w^{t+1}\|^2 \leq \ell R^2 \quad \text{from (3)}$$

$\Rightarrow \|w^{t+1}\|^2 \leq \ell R^2$

$\Rightarrow \boxed{\ell \leq \frac{R^2}{\gamma^2}}$ ← Radius Margin Bound

Since $\gamma^2 > 0$, no. of mistakes is finite

∴ **PERCEPTRON CONVERGES**

Sigmoid Functions for Modeling Class Probabilities

Linear Separability in terms of Probability:

$$P(y = 1/x) = 1 \quad \text{if } w^T x \geq 0$$
$$= 0 \quad \text{otherwise}$$

Can we somehow relax this assumption? Can we model probabilities differently?

Start with a simple linear model.

$$z = w^T x \leftarrow \text{feature } x \in \mathbb{R}^d$$

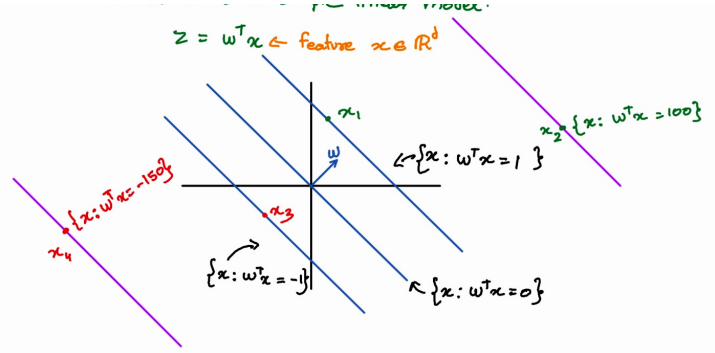


Figure 10. A 2D plot showing a line through the origin (normal vector w) with several points x_1, x_2, x_3, x_4 marked on either side, labeled $\{x : w^T x = -15\}$, $\{x : w^T x = -5\}$, $\{x : w^T x = 0\}$, $\{x : w^T x = 1\}$, $\{x : w^T x = 100\}$, illustrating the decision boundary.

How to label $\{x_1, x_2, x_3, x_4\}$? We are going to attach probability for the label to be +1 or -1 to each point.

How to decide the probability? By high or low the "score" of each point is. The score will be $z = w^T x$, $z \in [-\infty, \infty]$

- Larger the score ($z = w^T x$) more the probability of +1

Nice versa

Link Function

- $g(z) = 0$ if $z = 0$ (point lying on the separator)
- $g(z) \to +1$ if $z \to \infty$
- $g(z) \to -1$ if $z \to -\infty$

$$g(z) = \frac{1}{1 + e^{-z}} \quad \text{(Sigmoid/logistic function)}$$

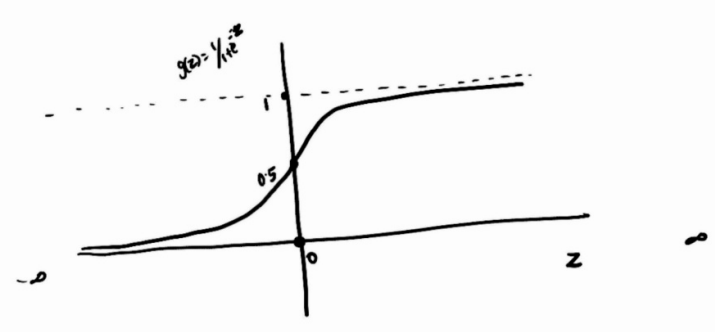


Figure 11. A sigmoid curve labeled $g(z) = 1/e^{z^2}$, with asymptotes at $y = 0$ and $y = 1$, crossing through $(0, 0.5)$, and increasing steeply from left to right.

Logistic Regression

$$P(y = 1/x) = \frac{1}{1 + e^{-w^T x}} \quad \text{Dataset: } \{(x_1, y_1), \dots, (x_n, y_n)\}$$

How to find w: **Maximum Likelihood**

$$L(w; \text{Data}) = \prod_{i=1}^{n} (g(w^T x_i))^{y_i} (1 - g(w^T x_i))^{(1-y_i)}$$

$$\log L(w; \text{Data}) = \sum_{i=1}^{n} y_i \log(g(w^T x_i)) + (1 - y_i) \log(1 - g(w^T x_i))$$

$$= \sum_{i=1}^{n} y_i \log\left(\frac{1}{1 + e^{-w^T x_i}}\right) + (1 - y_i) \log\left(\frac{e^{-w^T x_i}}{1 + e^{-w^T x_i}}\right)$$

$$\log L(w) = \sum_{i=1}^{n} \left[(1 - y_i)(-w^T x_i) - \log(1 + e^{-w^T x_i})\right]$$

Goal: $\max_w \sum_{i=1}^{n} (1 - y_i)(-w^T x_i) - \log(1 + e^{-w^T x_i})$

No closed form expression — can perform gradient descent (ascent)

$$\nabla \log L(w) = \left[(1-y_i)(-x_i) - \left(\frac{e^{-w^T x_i}}{1+e^{-w^T x_i}} \right) (-x_i) \right]$$

$$= \left[-x_i + y_i x_i + x_i \frac{e^{-w^T x_i}}{1+e^{-w^T x_i}} \right]$$

$$= \sum_{i=1}^{n} \left[-x_i + y_i x_i + x_i \frac{e}{1+e^{-w^T x_i}} \right]$$

$$= \sum_{i=1}^{n} \left[y_i x_i - x_i \left(\frac{1}{1+e^{-w^T x_i}} \right) \right]$$

$$\nabla \log L(w) = \sum_{i=1}^{n} \left[x_i \left(y_i - \frac{1}{1+e^{-w^T x_i}} \right) \right]$$

Gradient update rule

$$w_{t+1} = w_t + \eta_t \nabla \log L(w)$$

$$= w_t + \eta_t \left(\sum_{i=1}^{n} \left[x_i (y_i - \frac{1}{1+e^{-w_t^T x_i}}) \right] \right)$$

[Diagram: A green box containing x_i, an arrow to y_i, an arrow to $\theta_i \in \mathbb{R}^d$, an arrow back to the green box via $g(w_t^T x_i)$, with a label $\{0,1\}$ near y_i.]

Regularized Version

$$\min_w \sum_{i=1}^{n} \left[\log(1+e^{-w^T x_i}) + y_i x_i (1-y_i) \right] + \frac{\lambda}{2} \|w\|^2$$

cross validate hyperparameters. Regularized.